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LAB-3 Architecture Justification

Multi-Cloud Asset Cost Optimization Portal

Architecture Selection: I chose Layered Architecture for the Multi-Cloud Asset Cost Optimization Portal.

Architectural Choice: The system is organized into a Presentation Layer, Business Layer, and Data Layer. The FinOps Analyst/Cloud Administrator Dashboard belongs to the Presentation Layer. The Billing Ingestion Service, Cost Analysis Engine, Budget & Alert Manager, Notification Service, Report Generator, and Access Control Service belong to the Business Layer. The Billing Data Store belongs to the Data Layer.

Reason 1 – Separation of responsibilities: Ingesting and parsing raw billing CSVs is kept separate from cost analysis and idle-asset detection, which is kept separate again from budget-threshold checking and alert dispatch. This keeps the ingestion format (AWS/Azure/GCP CSV schema differences) isolated from the business rules that decide what counts as idle or over-budget.

Reason 2 – Maintainable billing configuration: Keeping billing records, budget thresholds, and flagged-asset history in a separate data layer gives the Cost Analysis Engine, Budget & Alert Manager, and Report Generator one consistent place to read and write cost data, instead of each service managing its own copy.

Security Advantage: Routing every threshold edit and alert dismissal through a dedicated Access Control Service enforces role-based access control at a single checkpoint, so only the Cloud Administrator role can modify budgets or dismiss breach alerts, and billing data in the Data Layer stays encrypted at rest (AES-256).

Performance Benefit: The Cost Analysis Engine can parse multi-gigabyte billing files and update the cost dashboard directly against the Data Layer without waiting on notification or reporting logic, helping the portal meet the under-2-minute update target even under peak load.

Key Components: Billing Ingestion Service; Cost Analysis Engine; Budget & Alert Manager; Notification Service; Report Generator; Access Control Service; Billing Data Store.

Key Interfaces: IIngestedCostData; IIdleAssetFeed; INotify; IAccessControl; IDataStore (Data Interface); IExportReport.